Project Initialization and Planning Phase

Date	15 July 2026
Team ID	SWTID-2026-3798
Project Title	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The AI-Powered Flood Prediction System is developed to predict floods before they occur. The system analyzes rainfall and water-level data using machine learning algorithms. It helps authorities and citizens receive early warnings and reduce flood-related damage.

Project Overview	
Objective	The primary objective is to develop an intelligent system that predicts floods using machine learning techniques and environmental data analysis.
Scope	The project comprehensively analyzes rainfall, water levels, and historical flood data to generate flood risk predictions and alerts.
Problem Statement	
Description	: Existing flood warning systems may provide late alerts and less accurate predictions.
Impact	: Increased risk to human lives, property damage, delayed emergency response, and economic losses.
Proposed Solution	
Approach	: Train machine learning models using historical and real-time environmental data to predict flood risks accurately.
Key Features	: Real-time flood prediction, Risk classification, Early warning alerts, Data visualization dashboard, Historical trend analysis.

Dataset requirements :

The project uses historical flood data in CSV format, along with weather and environmental data such as rainfall records and water-level measurements, to analyze and predict flood conditions. .

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Processing Unit	Intel i5 / Ryzen 5 or higher
Memory	RAM specifications	8 GB
Storage	Data Storage	20 GB Free Space
Software		
Programming Language	Development	Python
IDE	Development Environment	VS Code / Jupyter Notebook
Framework	Web Framework	Flask
Libraries	Data Processing & ML	Pandas, NumPy, Scikit-learn, Matplotlib